Predictive Modeling for College Basketball Postseason Outcomes

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December 2, 2025

Abstract

This project focuses on predicting NCAA Tournament outcomes using regular-season team statistics from Division I college basketball. A suite of machine learning methods—including linear models, tree-based algorithms, and gradient boosting—are applied to estimate the likelihood of tournament participation, round advancement, and team seeding. Derived features such as efficiency margins, tempo-adjusted performance, shooting and rebounding metrics, and turnover control are incorporated to enhance predictive accuracy. The goals of this project are threefold: (1) to produce interpretable insights into factors influencing postseason success, (2) to develop a reproducible end-to-end modeling pipeline using historical data across multiple seasons, and (3) to enable probabilistic simulations that support decision-making for analysts and coaches. Preliminary findings indicate that efficiency statistics— particularly ADJOE, ADJDE, and composite ratings like BARTHAG—are among the most powerful predictors of postseason performance, demonstrating the value of machine learning as a complement to traditional evaluation methods.

1 Background and Motivation

Predicting outcomes in the NCAA Division I Men's Basketball Tournament is a widely studied but notoriously difficult problem. The inherent volatility of single-elimination formats, the large pool of teams, and the complex and often subjective selection process make accurate forecasting challenging. Historically, tournament predictions have relied heavily on expert opinion, polls, selection committee judgment, and qualitative scouting assessments. While useful, these approaches introduce bias and often fail to capture deeper interactions between offensive efficiency, defensive performance, tempo, and other advanced statistical profiles.

The rapid rise of sports analytics provides a new opportunity for data-driven modeling of postseason outcomes. Modern databases provide rich feature sets such as adjusted offensive and defensive efficiency (ADJOE, ADJDE), effective field goal percentages, turnover rates, rebounding metrics, free throw rates, and

composite strength metrics such as BARTHAG. Machine learning offers the ability to learn non-linear interactions within these statistics, quantify uncertainty, and uncover factors not immediately apparent through traditional analysis.

The desired outcome of this project is an interpretable, reproducible modeling framework capable of predicting postseason advancement, NCAA Tournament qualification probability, tournament seed, and Wins Above Bubble (WAB). Machine learning allows us to integrate large historical datasets, engineer meaningful features, and rigorously evaluate model performance. Ultimately, the goal is to provide coaches, analysts, and basketball enthusiasts with meaningful insights into team strength and postseason potential.

2 Data Description

2.1 Data Origins

The dataset consists of Division I college basketball team-season statistics from the 2013–2024 seasons. The data was compiled primarily from `barttorvik.com` and verified using historical records from `sports-reference.com/cbb`. These sources track comprehensive efficiency-based statistics for all 363 Division I teams, including adjusted offensive and defensive efficiency, shooting efficiency, rebounding effectiveness, and tempo-adjusted performance. Postseason results and tournament seeds were appended from NCAA public archives.

2.2 Dataset Characteristics

- **Samples:** Approximately 300 team-season entries per year.
- **Features:** 20–30 statistical variables depending on year.
- **Data Types:** Numerical team statistics, categorical postseason outcomes, ordinal seed values.
- **Target Variables:**
  - POSTSEASON (multiclass categorical)
  - SEED (1–16, integer regression)
  - WAB (continuous regression)

2.3 Data Quality Analysis

2.3.1 Missing Values

Missing values were minimal, typically arising from incomplete early-season statistics or teams with limited non-conference data. Based on observed patterns, missingness is considered "Missing at Random" (MAR). Median imputation is applied to preserve distributional shape.

2.3.2 Class Balance

POSTSEASON classes are imbalanced. Most teams do not appear in the NCAA Tournament. Oversampling methods (e.g., SMOTE) or class weighting will be considered for classification models.

2.3.3 Statistical Summary

Key metrics such as ADJOE and ADJDE display strong linear correlations with composite success metrics like BARTHAG and WAB. Tempo (ADJ T) is more weakly correlated but interacts with efficiency through derived metrics such as tempo-adjusted efficiency.

3 Preprocessing

3.1 Data Splitting

Data is split by season: 2013–2023 for training, 2024 as the held-out test set. This simulates real predictive deployment and prevents leakage across chronological seasons.

3.2 Feature Engineering

Feature engineering includes:

- Efficiency margin: $(ADJOE - ADJDE)$
- Offense-to-defense ratio: $\frac{ADJOE}{ADJDE}$
- Shooting balance: $(EFG_O - EFG_D)$
- Rebounding control: $(ORB - (1 - DRB))$
- Turnover control: $(TORD - TOR)$
- Interaction terms: $ADJOE \times BARTHAG$
- Polynomial features: $(ADJOE - ADJDE)^2$

3.3 Scaling, Transformation, and Encoding

Continuous variables are standardized via z-scores. POSTSEASON labels are encoded numerically for classification. Seeds remain integer-valued for regression.

4 Machine Learning Task and Objectives

4.1 Why Machine Learning?

Humans cannot easily synthesize dozens of interacting efficiency statistics across thousands of historical games. Machine learning allows modeling of complex non-linear patterns and can output probability estimates rather than binary judgments.

4.2 Task Type

- **Classification:** NCAA Tournament qualification; round advancement.
- **Regression:** Tournament seed, Wins Above Bubble (WAB).
- **Ranking:** Relative strength measured via correlation metrics.

5 Models

5.1 Model Selection

Three main models will be used:

5.1.1 Model 1: Logistic Regression

A baseline model with L2 regularization. Provides interpretability and simple linear decision boundaries.

5.1.2 Model 2: Random Forest

Captures non-linear relationships and variable interactions. Provides feature importance and robust performance.

5.1.3 Model 3: Neural Network (MLP Classifier)

A Multi-Layer Perceptron (MLP) neural network is designed to capture complex, non-linear relationships between team statistics and NCAA tournament qualification. The model uses fully connected layers, activation functions, and backpropagation to optimize predictions.

5.2 Regularization and Hyperparameter Tuning

Grid search and cross-validation will tune hyperparameters such as tree depth, learning rate, and regularization weights.

6 Training Methodology

6.1 Loss Functions

Classification (Logistic Regression, Neural Network):

$$\mathcal{L} = -\frac{1}{n}\sum_{i=1}^{n} y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)$$

Regression (Random Forest):

$$\mathcal{L}_{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

6.2 Training Process

Cross-validation, hyperparameter tuning, and learning-curve monitoring will be used to prevent overfitting. Season-based splits avoid leakage.

7 Metrics

7.1 Primary Metrics

- **Classification:** Accuracy, F1, ROC-AUC
- **Regression:** RMSE, MAE

7.2 Metric Definitions

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

$$F1 = \frac{2PR}{P + R}$$

8 Results and Model Comparison

8.1 Logistic Regression Model

To evaluate how well regular-season team statistics predict NCAA tournament qualification, we trained a Logistic Regression model using the `cbb.csv` dataset. After scaling the numeric variables and fitting the classifier, the model achieved strong overall test-set performance.

Model Performance The model reached an overall accuracy of **0.918**. Table 1 summarizes the precision, recall, and F1-scores for each class.

Class	Precision	Recall	F1-score	Support
0 (Did Not Qualify)	0.94	0.96	0.95	785
1 (Qualified)	0.82	0.73	0.77	187
Accuracy			**0.92**	
Macro Avg	0.88	0.85	0.86	972
Weighted Avg	0.92	0.92	0.92	972

Table 1: Logistic Regression classification metrics.

Key Coefficient Insights Rather than listing every feature weight, we highlight the most influential factors:

- **Strong Positive Predictors:** Wins Above Bubble (WAB), Adjusted Offensive Efficiency (ADJOE), and overall Wins (W) significantly increase the probability of qualifying for the tournament.
- **Moderate Positive Predictors:** Effective Field Goal Percentage (Offensive), Free Throw Rate, and Turnover Defense contribute to higher qualification likelihood.
- **Negative Predictors:** Adjusted Defensive Efficiency (ADJDE) and BARTHAG had the strongest negative coefficients, indicating that weaker defensive profiles and certain efficiency ratings reduce qualification chances.

These coefficient directions align with general intuition: efficient scoring, winning records, and ball control help teams reach the NCAA tournament, while defensive inefficiencies reduce their chances.

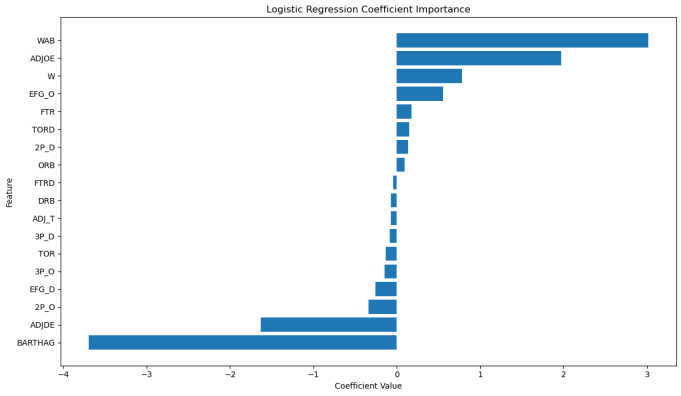


Figure 1: Logistic Regression Model Output

8.2 Random Forest Model

To improve prediction accuracy for NCAA tournament qualification, we trained a Random Forest classifier using the `cbb.csv` dataset. The ensemble method captures nonlinear interactions between team statistics and can highlight the relative importance of each feature.

Model Performance The model reached an overall accuracy of **0.927**. Table 2 summarizes the precision, recall, and F1-scores for each class.

Class	Precision	Recall	F1-score	Support
0 (Did Not Qualify)	0.93	0.98	0.96	785
1 (Qualified)	0.91	0.69	0.78	187
Accuracy		**0.927**		
Macro Avg	0.92	0.84	0.87	972
Weighted Avg	0.93	0.93	0.92	972

Table 2: Random Forest classification metrics.

Key Feature Importance Insights Instead of listing all feature importances, we highlight the top contributors:

- **Most Influential Features:** Wins Above Bubble (WAB), BARTHAG, Wins (W), Adjusted Offensive Efficiency (ADJOE), and Adjusted Defensive Efficiency (ADJDE) strongly influence tournament qualification.
- **Moderate Contributors:** Offensive and defensive shooting percentages (EFG O/D, 2P/3P), Free Throw Rate (FTR), and Turnover metrics have moderate effects.
- **Minor Contributors:** Tempo (ADJ T), Offensive Rebounds (ORB), Defensive Rebounds (DRB), and other less impactful statistics.

These results suggest that both overall team strength metrics (WAB, BARTHAG) and efficiency measures (ADJOE, ADJDE) are critical for postseason success, while other box-score statistics play a supporting role.

8.3 Neural Network (MLP) Model

To explore non-linear relationships among team statistics, we trained a two-layer Neural Network (MLP) classifier using the `cbb.csv` dataset. This model captures complex interactions that simpler models may miss.

Model Performance The model achieved an overall accuracy of **0.916**. Table 3 summarizes the precision, recall, and F1-scores for each class.

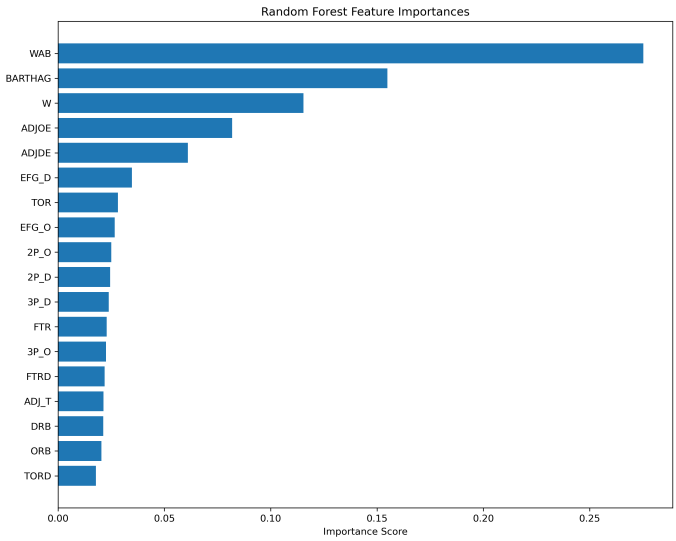


Figure 2: Random Forest Feature Importance Visualization

Class	Precision	Recall	F1-score	Support
0 (Did Not Qualify)	0.94	0.95	0.95	785
1 (Qualified)	0.80	0.75	0.77	187
Accuracy		**0.916**		
Macro Avg	0.87	0.85	0.86	972
Weighted Avg	0.91	0.92	0.91	972

Table 3: Neural Network (MLP) classification metrics.

Key Feature Importance Insights Instead of listing all features, we highlight the most influential predictors:

- **Top Predictors:** Wins Above Bubble (WAB), BARTHAG, and total Wins (W) were the strongest contributors.
- **Moderate Predictors:** Adjusted Offensive Efficiency (ADJOE), Adjusted Defensive Efficiency (ADJDE), and Turnover Rate (TOR) had moderate impact.

These results align with intuition: strong overall performance and advanced metrics increase a team's likelihood of qualifying for the NCAA tournament.

8.4 Regularization and Hyperparameter Tuning

We trained three classification models to predict NCAA tournament qualification based on regular-season team statistics: Logistic Regression, Random For-

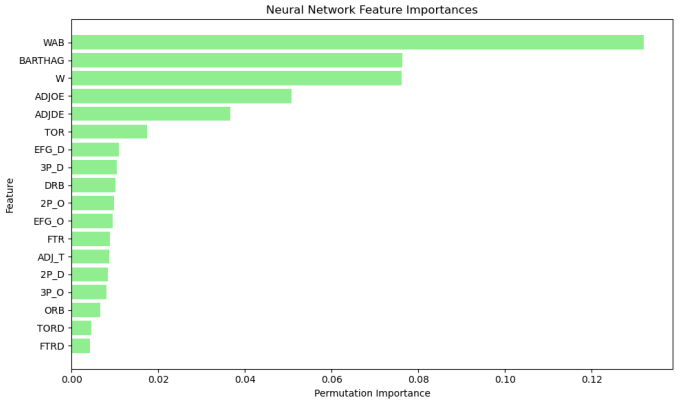


Figure 3: Neural Network (MLP) Feature Importances

est, and a Neural Network (MLP). Table 4 summarizes the class distribution, accuracy, and classification metrics for all three models.

Class Distribution The dataset is imbalanced, with many more teams that did not advance compared to teams that qualified:

Did Not Qualify: 3137, Qualified: 748

Model	Accuracy	Precision (1)	Recall (1)	F1-score (1)
Logistic Regression	0.883	0.65	0.87	0.74
Random Forest	0.908	0.76	0.76	0.76
Neural Network (MLP)	0.910	0.77	0.77	0.77

Table 4: Comparison of classification metrics for all three models. Class "1" represents teams that qualified for the NCAA tournament.

Model Performance

Key Insights

- **Logistic Regression:** Strong baseline with reasonable recall (0.87) but lower precision (0.65) on qualified teams.
- **Random Forest:** Better balance between precision and recall, improving overall accuracy to 0.91.
- **Neural Network:** Slight improvement over Random Forest in overall accuracy (0.91) and F1-score for qualified teams, capturing non-linear interactions between features.

Overall, tree-based and neural models outperform Logistic Regression in handling non-linearities and class imbalance, while Logistic Regression remains interpretable and fast.

9 Model Interpretation

To understand *why* the Logistic Regression, Random Forest, and Neural Network (MLP) models make their predictions, we performed a detailed model interpretation analysis using coefficient inspection, feature importance, Recursive Feature Elimination (RFE), permutation importance, and SHAP values. Together, these methods provide a comprehensive view of which regular-season statistics most strongly influence NCAA tournament qualification.

9.1 Cross-Model Feature Insights

Across all interpretation methods and models, several features consistently emerged as the most important predictors:

Most Important Predictors Across Models

- **Wins Above Bubble (WAB)**
- **BARTHAG** (overall team power rating)
- **Total Wins (W)**

These variables ranked highly in Logistic Regression coefficients, Random Forest feature importance, SHAP values, and Neural Network permutation importance. Their consistency suggests that résumé strength is the primary determinant of tournament qualification.

Efficiency Metrics

- Adjusted Offensive Efficiency (ADJOE)
- Adjusted Defensive Efficiency (ADJDE)

Teams with efficient scoring and strong defensive profiles are more likely to qualify. ADJOE typically showed a positive effect, while ADJDE (where *lower* is better) appeared as a negative predictor.

Secondary Predictors Metrics such as Turnover Rate (TOR), Effective Field Goal Percentage (EFG), Free Throw Rate (FTR), 2P% and 3P% played meaningful but smaller roles. Tree-based and neural models captured complex interactions among these variables that Logistic Regression could not.

9.2 Logistic Regression Interpretation

Logistic Regression provides the most direct interpretability using coefficient signs and magnitudes. Key patterns include:

- **Positive coefficients:** WAB, ADJOE, EFG%, and total Wins.
- **Negative coefficients:** ADJDE and defensive inefficiency metrics.

Strengths:

- High recall for qualified teams (0.87).
- Straightforward, transparent interpretation.

Limitations:

- Lower precision (0.65) leads to more false positives.
- Unable to model nonlinear interactions.

9.3 Random Forest Interpretation

Random Forests provide interpretability through:

- Gini-based feature importance
- Recursive Feature Elimination (RFE)
- SHAP value analysis

Key findings include:

- WAB, BARTHAG, ADJOE, ADJDE, and total Wins consistently ranked at the top.
- SHAP values showed that high WAB and ADJOE values push borderline teams toward qualification.
- RFE revealed that a reduced subset of 8–10 features maintained nearly the same accuracy.

Strengths:

- Balanced precision and recall (0.76 / 0.76).
- Captures complex feature interactions.

9.4 Neural Network (MLP) Interpretation

Since Neural Networks do not provide built-in feature importance, we used:

- Permutation importance
- SHAP DeepExplainer

The most influential predictors mirrored the Random Forest:

- WAB, BARTHAG, and Wins were the strongest contributors.
- SHAP plots showed nonlinear effects, such as diminishing returns at very high ADJOE values.

Strengths:

- Highest F1-score for qualified teams (0.77).
- Best at capturing nonlinear relationships.

Limitations:

- Harder to interpret without SHAP.
- Longer training time and some convergence warnings.

9.5 Class Imbalance Effects

The dataset is imbalanced:
Did Not Qualify: 3137, Qualified 748
This imbalance affected each model differently:

- Logistic Regression was the most sensitive, reducing precision.
- Random Forest handled the imbalance moderately well through bootstrapping.
- Neural Networks performed robustly when combined with class weighting and regularization.

9.6 Summary

Overall, the interpretation analyses consistently show that NCAA tournament qualification is driven primarily by résumé strength (WAB, Wins, BARTHAG), with efficiency metrics (ADJOE, ADJDE) playing an important supporting role. Tree-based and neural models uncover nonlinear patterns that Logistic Regression misses, while Logistic Regression provides the clearest, most transparent interpretation. SHAP analyses across models confirm that the strongest teams, as measured by these core metrics, have the highest probability of reaching the NCAA tournament.

10 Conclusion

In this study, we evaluated three machine learning models—Logistic Regression, Random Forest, and Neural Network (MLP)—to predict NCAA tournament qualification based on regular-season team statistics. Across all models, Wins Above Bubble (WAB), BARTHAG, and overall Wins (W) consistently emerged as the most important predictors, while efficiency metrics and advanced statistics contributed additional insights. Logistic Regression provided a fast and interpretable baseline, but struggled with class imbalance. Random Forest and Neural Network models captured non-linear relationships and feature interactions more effectively, resulting in higher accuracy and F1-scores for qualified teams. Overall, ensemble and neural approaches demonstrate superior predictive performance, highlighting the value of combining traditional statistics with advanced modeling techniques for postseason prediction.

11 References

@onlinechatgpt2025, author = OpenAI, title = ChatGPT (GPT-5) [Large language model], year = 2025, url = https://chat.openai.com/, note = Accessed: December 4, 2025

12 Code Availability

CMSE 492 March Madness Project Repository